FPGA Beamformer

Stewart Nash

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Chapter 1

Adaptive Filtering

"Filters that adapt to a changing environment are discussed. The stochastic Wiener filtering and deterministic least-squares problems are set up, and the similarity between them is pointed out. First, a block-processing approach is taken in the solution to these two problems. Second, a fading memory (recursive) approach is taken. Then the application of these algorithms to the adaptive beamforming (ABF) problem is considered. The chapter appendix provides background material on conventional beamforming (CBF)."

1.0 Introduction

Numerous applications exist that require a linear filtering operation. Often, the nature of that filtering task is time-varying in some nondeterministic fashion owing to nonstationarity of the underlying time series. In such situations, a filter that can adapt to a changing environment is needed. Examples include adaptive noise canceling, line enhancing, frequency tracking, and channel equalization. Beyond the discussion here, additional material on adaptive filtering can be found in references 1 through 16.

1.1 The Stochastic Wiener Filtering and Deterministic Least Squares Problem

The problem of interest is described in Figure 11.1. The goal is to filter the time series $x[n]$ with a causal, finite impulse response (FIR) digital filter to yield an estimate of the time series $s[n]$. The error in that estimate is denoted $e[n]$

$$e[n] = s[n] - \hat{s}[n] = s[n] + \sum_{k=0}^{N} h_k x[n-k] \tag{1.1}$$

Written in matrix form, Eq. 11.1 becomes

$$e[n] = s[n] + [h_0 \, h_1 \, \ldots \, h_N] \begin{bmatrix} x[n] \\ x[n-1] \\ \vdots \\ x[n-N] \end{bmatrix} \tag{1.2}$$

or, in vector notation,

$$e[n] = s[n] + \mathbf{h}^T \mathbf{x} \tag{1.3}$$